

Raghav Senthil Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Built a *Python (FastAPI)* data pipeline that collects, cleans, transforms and delivers **600K+ debate-evidence documents** via *Elasticsearch*, returning results in **under 1 second** for a high school debate research SaaS platform
- Identified and resolved data quality issues, de-duplicating the **30%** of evidence documents that were identical
- Validated source URLs and filtered statistical outliers (**IQR** on word counts) to extract clean evidence segments
- Worked in an *Agile* team of **10 engineers** to keep data quality high and deliver search to **10K+ site visitors**

Product Development Intern | Jetson

January 2025 – February 2025

- Participated in testing and validating **3 experimental features** across **7 iterations of UI/UX testing**
- Created documentation of test findings in a **5-page report** with strategies to optimize user engagement for the CEO
- Validated a redesigned *Figma* onboarding flow, cutting it from **12 to 5 pages** through **A/B testing** with 30 users

Data Analytics Intern | STEM E

June 2024 – July 2024

- Analyzed data from **200+ TikTok posts** over **3 years** with *Python (pandas)*, *Power BI* and *Microsoft Excel*
- Used the data analysis to identify that “science fun facts” drove **69.3%** of views from only **7%** of all content
- Delivered measurable value with a “fun facts” strategy the Executive Director adopted, lifting engagement **24%**

PROJECTS

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Wrote *SQL* queries on a relational *PostgreSQL* database to retrieve papers by field, year and citation count
- Built a dual-API data pipeline that collects papers from *OpenAlex* and backfills metadata from *Semantic Scholar*
- Integrated the *Claude API* and *Supernemory* into a **RAG** agent that retrieves session-scoped vector data
- Delivered data visualization of a paper’s outbound citations across **26 research fields** in a *React Flow* graph

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Built a *Python FastAPI* back end that pulls live water, waste and energy readings from all **3 edge AI tools**
- Flagged data anomalies with **4 Random Forest models** and delivered actionable steps via a *Gemma* Voice LLM
- Created **3 edge AI tools** that collect data on a school’s energy, water and food consumption for anomaly detection
- Led a team of 5 to deliver the data product from ideation to presentation. **USAII Global Hackathon Finalist**

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Collected **15,000** truck-listing images and cleaned the dataset by de-duplicating listings and filtering non-USD prices
- Tested and validated the extraction, fixing each wrong prediction until **10 consecutive images** came back right
- Transformed truck photos into structured condition, brand, model and year data with a *Claude Sonnet 4.6* agent
- Trained a *quantile regression* pricing model on extracted data and validated **87% accuracy** on **30 samples**

EcoBin | [Research Paper](#) | [Source Code](#)

March 2026 – May 2026

- Built an automated synthetic data pipeline (*U2-Net* segmentation, alpha-compositing) that generates **9K** images
- Created documentation of the data and the model’s methodology in an *arXiv* paper, releasing **2 datasets** on *Kaggle*
- Tested and validated a two-stage *EfficientNetV2-S* classifier on **15K+ images**, reaching **96.13% accuracy**
- Validated a new smart bin prototype on an *Arduino Uno* that sorts each waste item in under **2 seconds** of latency

SKILLS

Programming Languages: SQL, Python, Java, JavaScript, C++, HTML/CSS

Frameworks/Platforms: PostgreSQL, MySQL, Supabase, Elasticsearch, FastAPI, AWS, Flask, OpenAlex API, React

DevTools: Git, GitHub, CI/CD, Docker, VS Code, Claude Code, Codex, Figma, Bun, Arduino IDE, Kaggle, Clang

Data & Analytics: pandas, NumPy, Microsoft Excel, Power BI, A/B Testing, Hootsuite

Concepts/Practices: Data Pipelines, Data Quality, Data Transformation, Data Validation, Relational Databases